

Longest Substring Without Repeating Characters

Title : Finding the Longest Substring Without Repeating Characters

Objective / Aim : To write a Java program to find the longest substring without repeating characters in a given string.

System Requirements:

Hardware:

- Standard PC/Laptop
- Core i3/i5 Processor or equivalent
- 4 GB/8 GB RAM

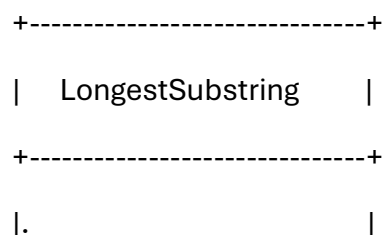
Software:

- Windows/Linux/macOS
- JDK 17 or JDK 21
- Eclipse / IntelliJ IDEA / NetBeans / VS Code
- Command Prompt or Terminal

Algorithm / Logic Description :

1. Start the program.
2. Read a string from the user.
3. Initialize variables to store the starting position and maximum length.
4. Check each character of the string.
5. If a character is repeated, move the starting position forward.
6. Continue until all characters are checked.
7. Store the longest substring found.
8. Display the length and the longest substring.
9. Stop the program.

UML Class Diagram:



```
+-----+
| + main(String[]) : void    |
| + findLongest(String) : String |
+-----+
```

Source Code:

```
import java.util.Scanner;

public class LongestSubstring {

    static String findLongest(String str){

        String longest = "";

        for (int l = 0; l < str.length(); l++) {

            String current = "";

            for (int j = l; j < str.length(); j++) {

                char ch = str.charAt(j);

                // Check whether character is already present
                if (current.indexOf(ch) != -1) {

                    break;

                }

                current = current + ch;

                // Store the longest substring

                if (current.length() > longest.length()) {

                    longest = current;

                }

            }

        }

        return longest;

    }

}
```

```

public static void main(String[] args) {

    Scanner sc = new Scanner(System.in);

    System.out.print("Enter a string: ");

    String str = sc.nextLine();

    String longest = findLongest(str);

    System.out.println("Length: " + longest.length());

    System.out.println("Longest substring: " + longest);

    sc.close();

}
}

```

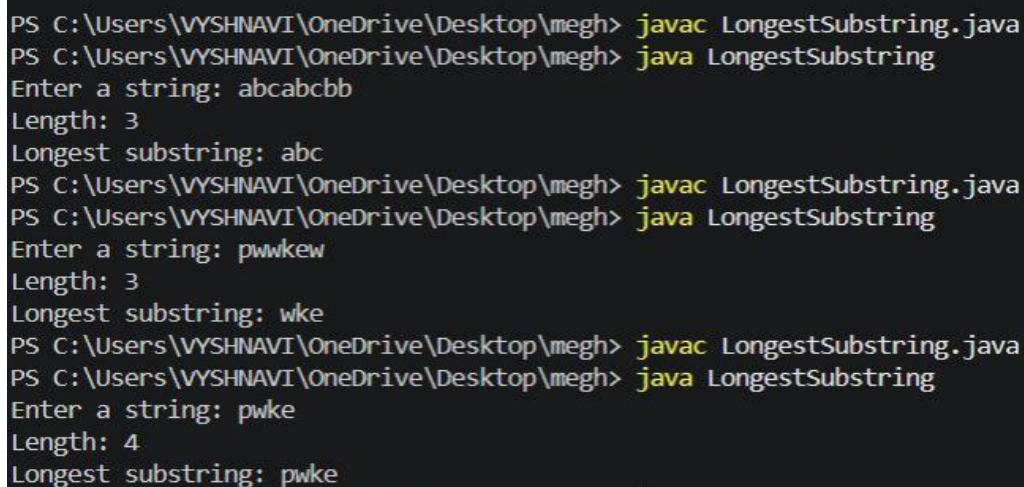
Compilation & Execution Commands :

Save the file as: LongestSubstring.java

Compile: Javac LongestSubstring.java

Run: Java LongestSubstring

Output screenshot :



```

PS C:\Users\VYSHNAVI\OneDrive\Desktop\megh> javac LongestSubstring.java
PS C:\Users\VYSHNAVI\OneDrive\Desktop\megh> java LongestSubstring
Enter a string: abcabcbb
Length: 3
Longest substring: abc
PS C:\Users\VYSHNAVI\OneDrive\Desktop\megh> javac LongestSubstring.java
PS C:\Users\VYSHNAVI\OneDrive\Desktop\megh> java LongestSubstring
Enter a string: pwwkew
Length: 3
Longest substring: wke
PS C:\Users\VYSHNAVI\OneDrive\Desktop\megh> javac LongestSubstring.java
PS C:\Users\VYSHNAVI\OneDrive\Desktop\megh> java LongestSubstring
Enter a string: pwke
Length: 4
Longest substring: pwke

```

Result:

Thus, the Java program to find the longest substring without repeating characters was successfully implemented and executed.